

Proctify : A Real Time Proctoring Solution

1. Introduction

Introduction: The "Proctify" project is a proctored exam tool designed to maintain the integrity of online exams. Developed for national and international level hackathons, it combines admin website, student website, browser-based, and machine learning components to create a secure exam environment.

2. Admin Website Features

2.1 Ongoing Exams

Ongoing Exams: The Ongoing Exams feature provides administrators with real-time monitoring capabilities during ongoing exams. Admins can view live webcam and screen feeds of individual students to detect any suspicious behavior or cheating attempts.

Proctor

Ongoing Exams

Create Exam

Generate Question

View Exams

View Questions

Student Details

Publish Result

Edit Profile

Exam

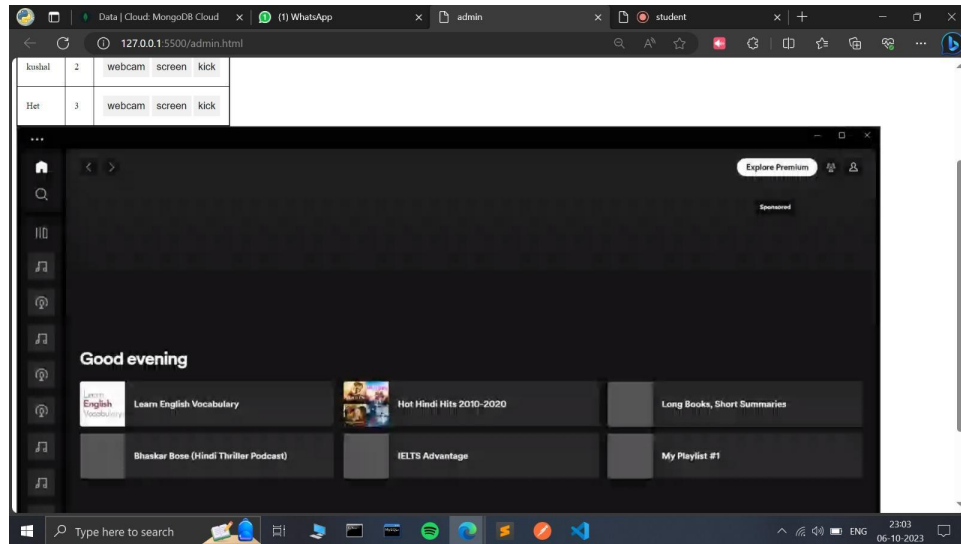
Live exam monitoring

ID	Student Name	CV Based	Mobile Detected	Suspicious Act	Noise Detected	Trust Score	Web Cam	Screen Share	Remove User
1	Jon Snow	35	5	5	5	5	Webcam	Screen	Remove
2	Cersei Lannister	42	5	5	5	5	Webcam	Screen	Remove
3	Jaime Lannister	45	5	5	5	5	Webcam	Screen	Remove
4	Anya Stark	16	5	5	5	5	Webcam	Screen	Remove
5	Daenerys Targ.	31	5	5	5	5	Webcam	Screen	Remove
6	Ever Melisandre	150	5	5	5	5	Webcam	Screen	Remove
7	Ferrara Clifford	44	5	5	5	5	Webcam	Screen	Remove
8	Rossini Frances	36	5	5	5	5	Webcam	Screen	Remove
9	Harvey Royle	65	5	5	5	5	Webcam	Screen	Remove

Rows per page: 1001-9 of 9



(Admin viewing webcam of a student)



(Admin viewing screen of a student)

2.2 Create Exam

Create Exam: This feature empowers administrators to create new exams and reschedule existing ones. It offers a user-friendly interface for specifying exam details, such as date, time, duration and more.

(Create New Exam)

The screenshot shows the 'Proctor' sidebar on the left with the 'Existing Exam' option highlighted. The main content area is titled 'Create from Existing Exam' and features an 'Exam Code' search bar. Below this are three input fields: 'Select Date' with a calendar icon, 'Start Time' with a clock icon, and 'End Time' with a clock icon. A blue 'CREATE EXAM' button is positioned at the bottom center.

(Reschedule Existing Exam)

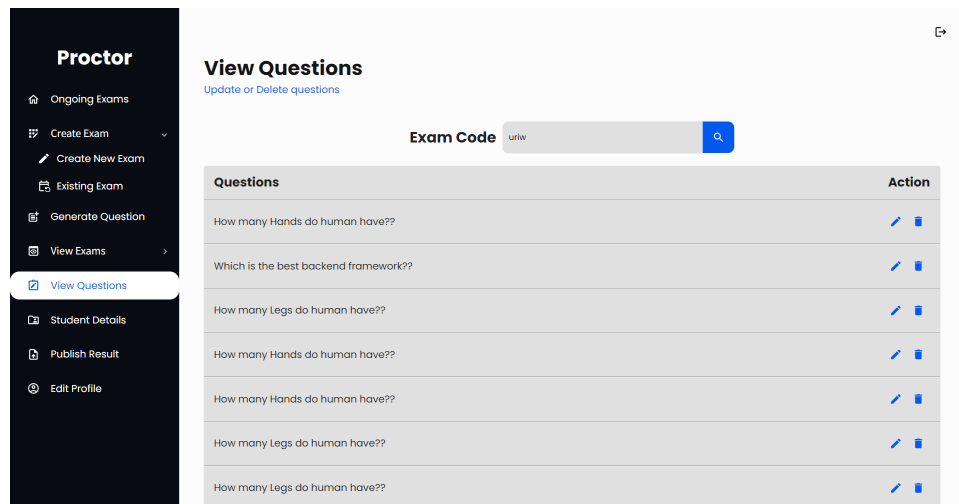
2.3 Generate Questions

Generate Questions: Admins can use this feature to create exam questions easily. It provides options for specifying question types (image question or simple question), answer options, and other parameters to ensure a well-rounded set of questions.

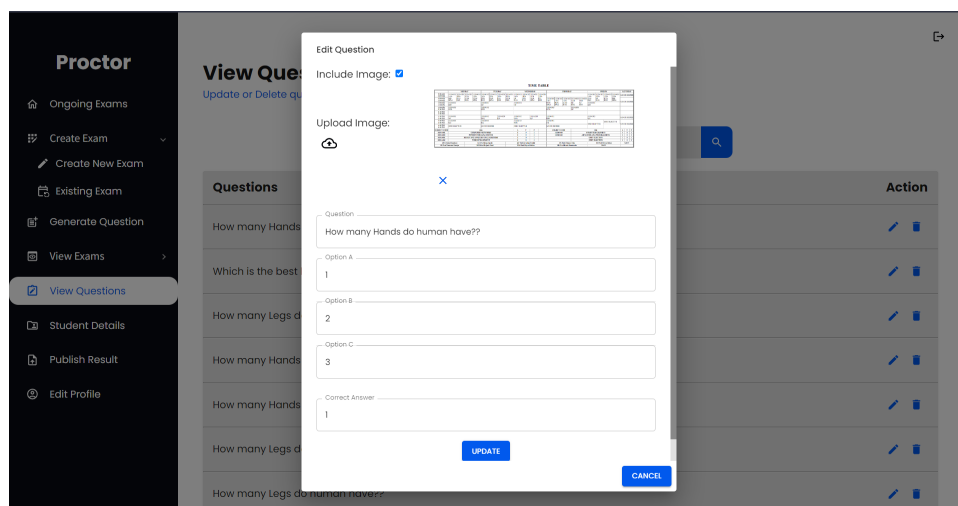
The screenshot shows the 'Proctor' sidebar on the left with the 'Generate Question' option highlighted. The main content area is titled 'Generate Question' with a 'Create Question' link. It includes an 'Exam Code' search bar with a magnifying glass icon and an 'Include Image' checkbox. Below these are four input fields: 'Question *', 'Choose No of Options' (a dropdown menu showing '2'), 'Option 1', 'Option 2', and 'Choose Correct Option' (a dropdown menu). A blue 'SUMMIT' button is located at the bottom center.

2.4 View Questions

View Questions: Admins can manage exam questions efficiently by viewing, editing, or deleting them. This feature simplifies question management and helps ensure the quality of exam content.



(View Questions)



(Edit Question)

2.5 View Exams

View Exams: Access to logs from previous exams is essential for auditing and analysis. This feature enables admins to review exam history, including student performance data and any anomalies. It also contains the feature to edit the date, time and more for upcoming exam.

Proctor

Ongoing Exams

Create Exam

Create New Exam

Existing Exam

Generate Question

View Exams

Edit New Exam

Past Exams

View Questions

Student Details

Publish Result

Edit Profile

Edit Exams

Manage exams and student details

Designing

Exam Date: 2023-10-07

Start Time: 00:00

End Time: 18:00

EDITDELETE

(Upcoming Exams)

Edit Exam

Edit exam details:

Exam Name *

Designing

Select Date

10/07/2023

Start Time

12:00 AM

End Time

07:00 PM

Duration (minutes) *

20

Negative Marking (%) *

4

Question Weights *

5

Exam Mode

Randomized student question formats?

Auto

Live

Yes

No

Edit student details:

Requirement

Age

Formal

Number

+

CANCELUPDATE

(Edit Upcoming Exam)

Proctor

Ongoing Exams

Create Exam

Generate Question

View Exams

Edit New Exam

Past Exams

View Questions

Student Details

Publish Result

Edit Profile

Past Exams

View logs of past exams

Bank Exam

Date: 2023-09-18T18:30:00.000Z

VIEW DETAILS

GCET

Date: 2023-09-26T18:30:00.000Z

VIEW DETAILS

GCET

Date: 2023-09-26T18:30:00.000Z

VIEW DETAILS

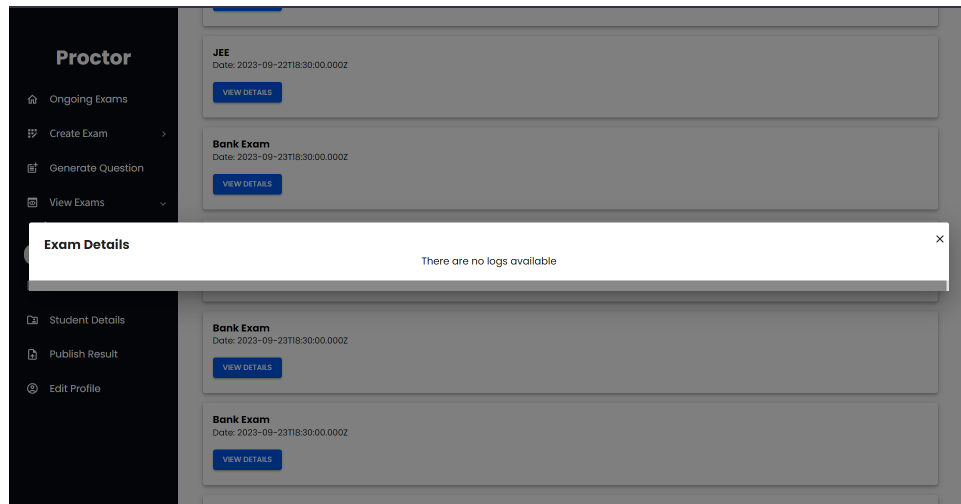
NEET

Date: 2023-09-22T18:30:00.000Z

VIEW DETAILS

JEE

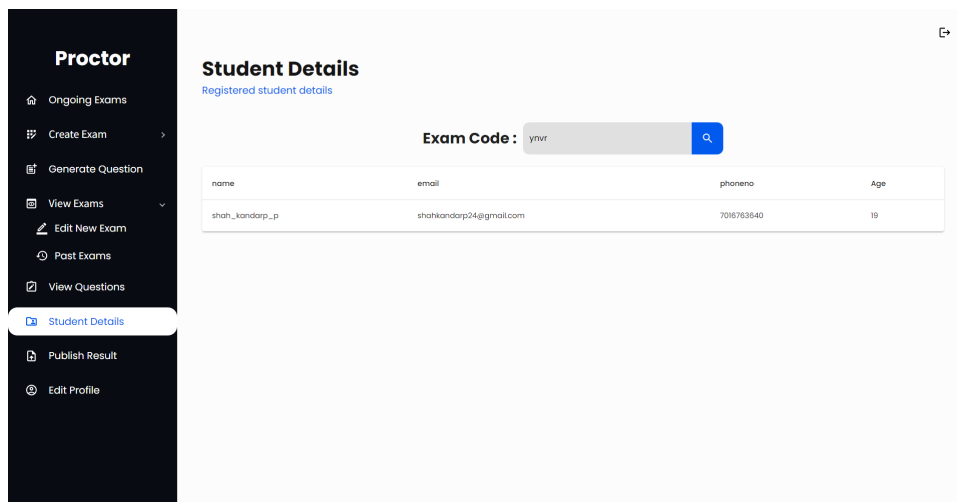
(Previous Exams)



(Logs of previous exams)

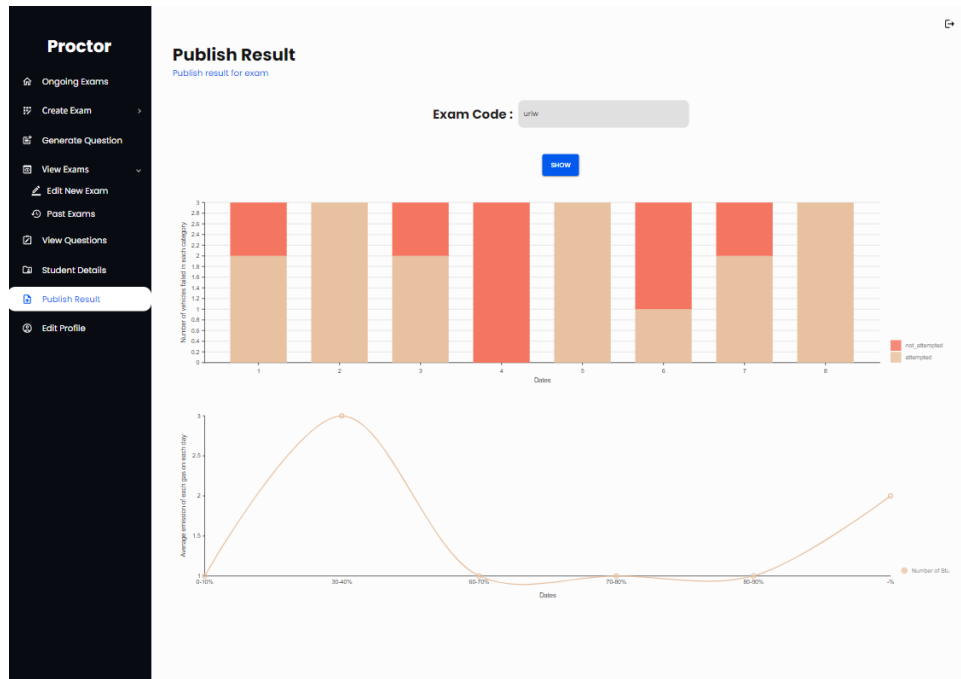
2.6 Student Details

Student Details: Admins can access comprehensive student registration details, facilitating exam administration and tracking of student participation.



2.7 Publish Result

Publish Result: Admins can release exam results and view detailed performance analytics. This feature provides insights into student performance and helps identify trends.



2.8 Edit Profile

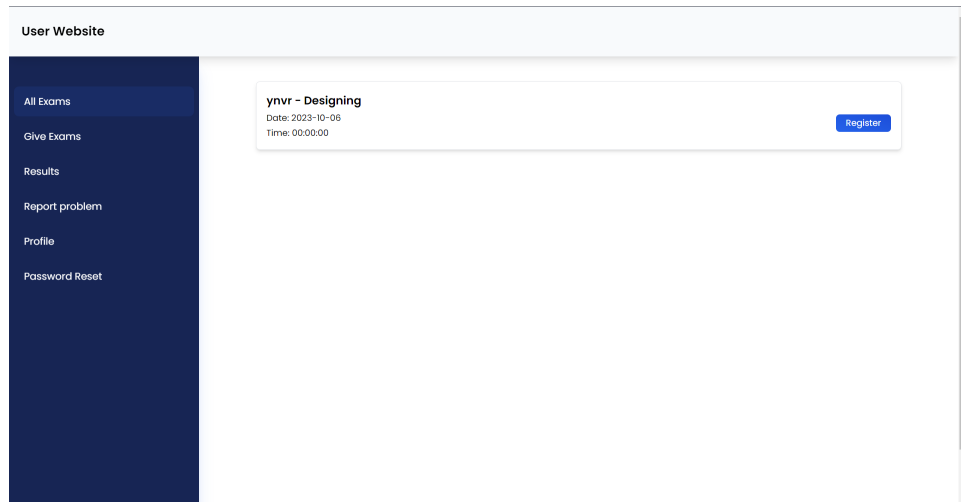
Edit Profile: Admins can keep their profiles up-to-date by editing their information easily through this feature.

The screenshot displays the 'Proctor' application interface for editing a profile. The sidebar menu is identical to the previous screenshot, with 'Edit Profile' highlighted. The main content area is titled 'Edit Profile' with the subtitle 'Update your profile'. It contains a form with three input fields: 'Name' (containing 'kandarp_shah'), 'Email' (containing 'shahikandarp24@gmail'), and 'Phone Number' (containing '123456789'). An 'UPDATE' button is located below the form.

3. Student Website Features

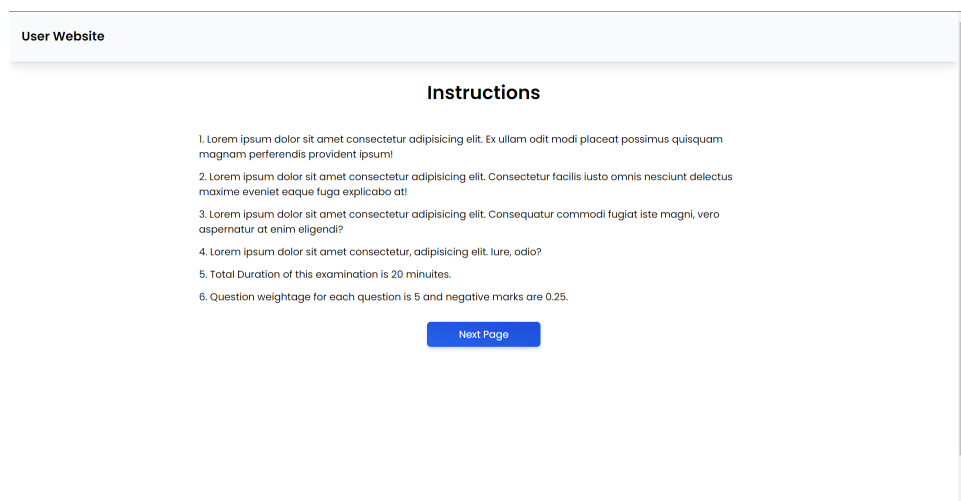
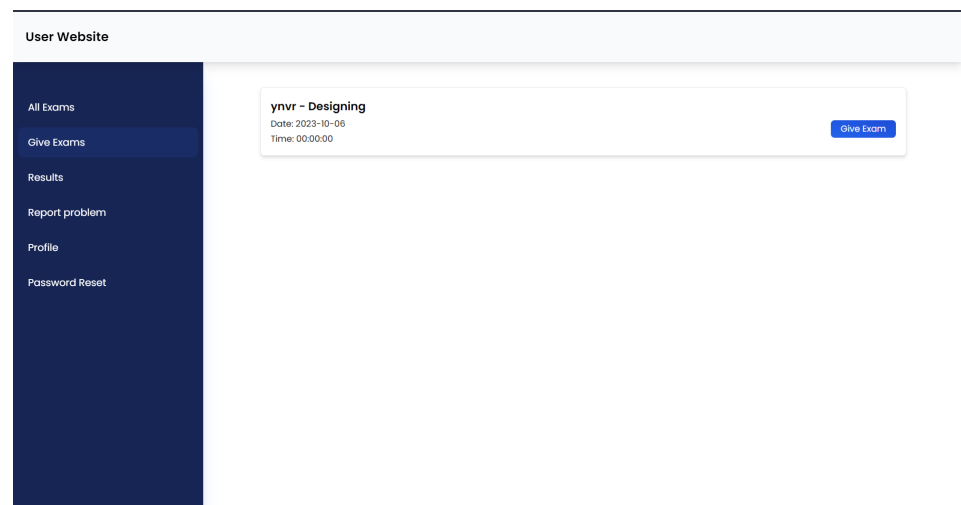
3.1 All Exams

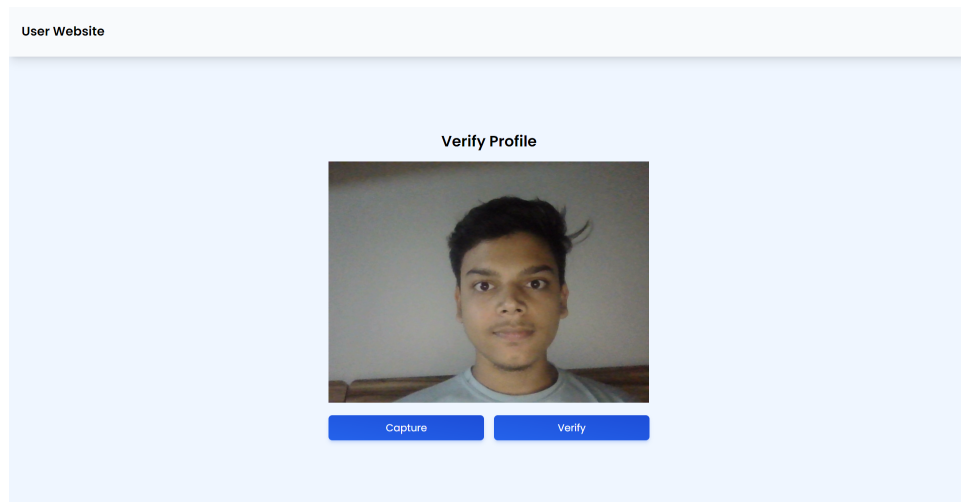
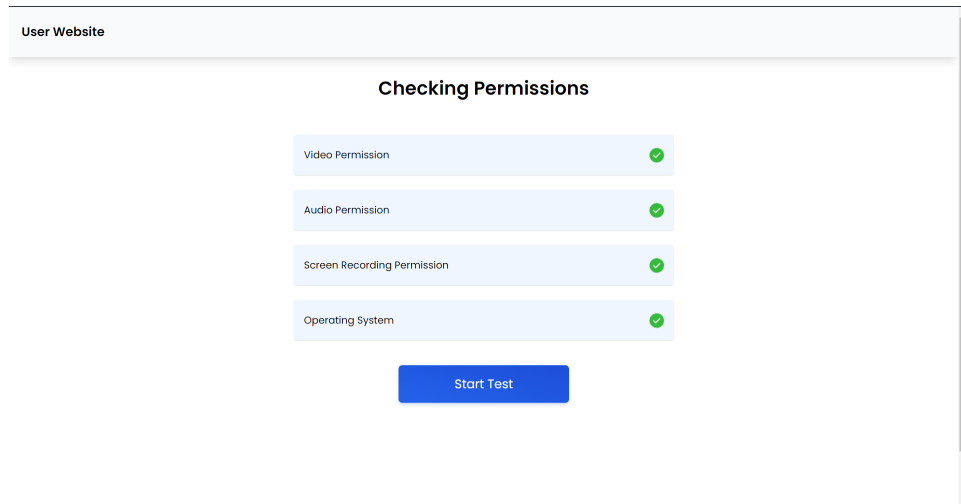
All Exams: Students can explore a list of upcoming exams and register for the ones they wish to take. Face verification is employed during registration to verify student identity.



3.2 Give Exams

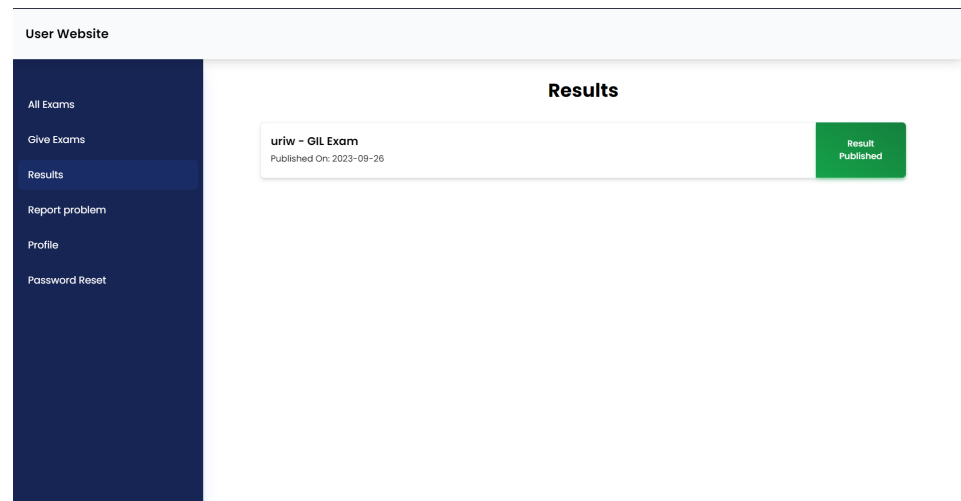
Give Exams: This feature allows students to access the exams they've registered for and begin taking them. Face verification is required at the start of each exam to prevent impersonation.





3.3 Results

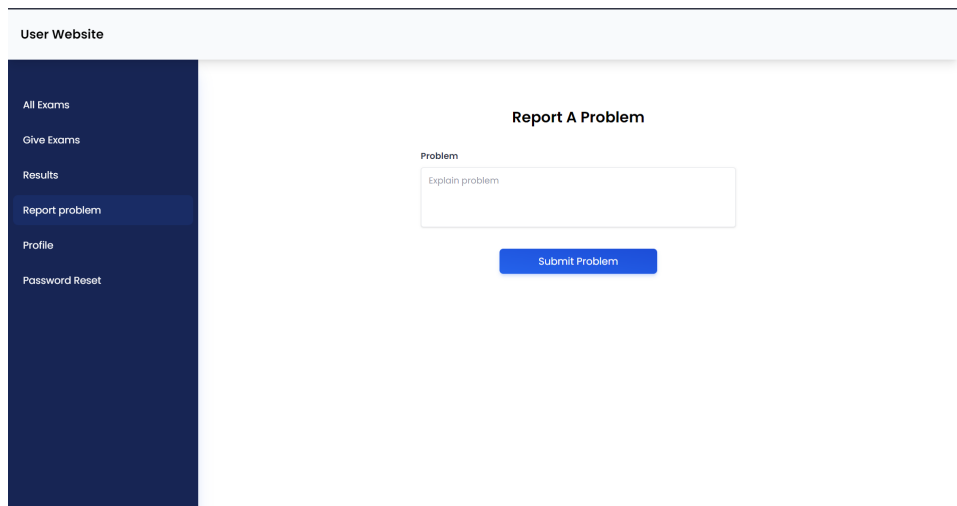
Results: The Results section displays student's exam results, providing a detailed breakdown of their performance.





3.4 Report Problem

Report Problem: Students can report any issues they encounter during exams or problems related to the software. This feature ensures a smooth exam experience by addressing issues promptly.



3.5 Edit Profile and Password Reset

Edit Profile and Password Reset: These user-friendly features enable students to manage their profiles and reset their passwords as needed.

User Website

My Profile 

Name: shah_kandarp_p
Email: shahkandarp24@gmail.com
Phone No: 7016763640

(Edit Profile)

User Website

Reset Password

Current Password

New Password

Confirm Password

[Change Password](#)

(Reset Password)

4. Self-Created Browser Features

4.1 Secure Exam Environment

Secure Exam Environment: The self-created browser is designed to create a secure environment for exams. It achieves this by blocking key combinations (e.g., Ctrl+C, Alt+Tab, Win+D) and detecting HDMI ports.

4.2 Blocking Unwanted Apps

Blocking Unwanted Apps: The browser prevents students from using unauthorized software during exams, further enhancing security.

5. Machine Learning Model Features

5.1 Cheating Detection

Cheating Detection: The machine learning model employs various techniques to detect cheating behaviors. These include cell phone usage detection, identification of suspicious behavior (e.g., looking away), noise detection, low-light condition detection, detection of multiple persons in the frame, and recognition of the presence of unauthorized materials like books.

6. Conclusion

Conclusion: In summary, the Proctify project is a powerful tool that enhances exam security and integrity. It offers a comprehensive set of features for administrators and students alike, backed by robust cheating detection capabilities.